

Avinash Malladi

malladiavinash3@gmail.com | (+91) 6281855895 | LinkedIn | GitHub | Credly

Summary

B.Tech Computer Science student with hands-on experience building **web applications, backend systems, REST APIs, and database-driven solutions** using **Python, JavaScript, SQL, Flask, MySQL, and SQLite**. Experienced with **Git, Docker, GitHub Actions, AWS, CI/CD, testing, and technical documentation**, with projects involving API-driven applications, real-time dashboards, debugging tools, and AI-powered software. Interested in **software engineering, backend development, full-stack technologies, cloud platforms, and building reliable, scalable applications**.

Education

B.Tech in Computer Science and Engineering	Aug 2023 – Present
SR University, Warangal	CGPA: 8.6 / 10
Intermediate (11th & 12th)	Aug 2021 – May 2023
Narayana Junior College, Hyderabad	92.9%
High School	2020
Krishnaveni Talent School, Mancherla	CGPA: 10 / 10

Technical Skills

Programming: Python, JavaScript, TypeScript, SQL, C
Backend & APIs: Node.js, Express.js, Flask, REST APIs, API Design, OOP
Frontend: React.js, HTML, CSS, JavaScript
Web & Databases: MySQL, SQLite
Software Engineering: Data Structures & Algorithms, Git, GitHub, Unit Testing, Integration Testing, Debugging, Code Analysis, Technical Documentation
Cloud & DevOps: AWS, Docker, GitHub Actions, CI/CD
Generative AI: RAG, LLM Applications, BM25, Information Retrieval, Agentic Workflows, Prompt Engineering, LangChain, SHAP, LIME, PDP, ICE
Machine Learning: Scikit-learn, TensorFlow, XGBoost, Feature Engineering, Model Evaluation, SMOTE

Projects

Smart Disaster Relief System

- Developed a **Flask-based web application** with **SQLite** for managing disaster reports, relief camps, resources, and victim allocation through backend-driven workflows.
- Implemented **risk classification, geolocation, resource-aware allocation**, and nearest-camp detection using OpenStreetMap Nominatim and rule-based decision logic.
- Built a real-time monitoring dashboard using **JavaScript, Leaflet.js, Chart.js, and heatmaps**, with periodic backend polling, risk alerts, geolocation, and map-based routing.

AI Code Debugger

- Developed an **AI-assisted Python debugging application** that analyzes source code to identify syntax, runtime, and logical errors.
- Implemented automated **error classification, explanations, and debugging suggestions** using Python code parsing, machine learning, and NLP techniques.

SR University AI Assistant – Agentic RAG Chatbot

- Developed an **AI-powered web application** for university policy and academic queries using RAG with **page-level source citations**.
- Implemented a custom **BM25 retrieval pipeline** with document routing, chunking, stemming, phrase matching, and reranking to improve retrieval relevance.
- Built an **agentic tool workflow** integrating handbook search, safe arithmetic calculation, and optional web search with fallback handling for tool and LLM failures.
- Created a **golden Q&A evaluation framework** measuring Hit Rate and MRR; containerized the application using **Docker** with CI/CD-ready automated evaluation.

Explainable AI-Powered Lung Cancer Prediction – Multi-Model ML Study *[Verify]*

- Co-authored and presented a research paper at the **IEEE ICICCS 2026 International Conference** on explainable machine learning for lung cancer prediction.
- Evaluated multiple ML models, achieving **98% accuracy using XGBoost**; applied **SHAP, LIME, PDP, and ICE** for model interpretation.

Certifications

Amazon Certified Cloud Practitioner – Amazon Skill Builder	May 2026	<i>[Verify]</i>
Microsoft Azure AI Fundamentals (AI-900)	Aug 2025	<i>[Verify]</i>
AWS Academy Data Engineering – AWS	Feb 2025	<i>[Verify]</i>
Python Full Stack – EduSkills	Dec 2024	<i>[Verify]</i>